OpenPrinting

Continuous Integration

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Continuous Integration

- Mostly Github Actions workflows used to validate/test software changes
- "Build" workflows to build projects on various OS's (Linux, macOS, Windows) and architectures (ARM, x86, etc.)
  - `./configure && make && make test`
- "CodeQL", "Coverity", "LGTM", and other static code analysis workflows to look for errors/issues
- "Docker" for making Docker images
- Also snapcraft, AppImage, Flatpak, etc. builders which are triggered by changes pushed to the corresponding repositories

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Github Actions

- Defined in YML files under ".github/workflows" in your repository
- Actions can be triggered by pushes to the repository or started manually
- Private data (access tokens, certificates, etc.) can be added to the environment that is passed
- Some services available under the "Actions" tab on the web site, others you need to setup by hand or through a third-party web site

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Example: PAPPL build.yml

```
name: Build

on:
  push:
    branches: [ master ]
  pull_request:
    branches: [ master ]

jobs:
  build-linux:

    runs-on: ubuntu-20.04

    steps:
    - uses: actions/checkout@v2
    - name: update build environment
      run: sudo apt-get update --fix-missing -y
    - name: install prerequisites
      run: sudo apt-get install -y avahi-daemon
 cppcheck libavahi-client-dev libcups2-dev
 libcupsimage2-dev libgnutls28-dev libjpeg-dev
 libpam-dev libpng-dev libusb-1.0-0-dev zlib1g-
 dev
    - name: configure
      env:
        CC: /usr/bin/gcc
      run: ./configure --enable-debug --enable-
 maintainer
    - name: make
      run: make
    - name: test
      env:
        ASAN_OPTIONS: leak_check_at_exit=false
      run: make test || cat testsuite/
testpappl.log
    - name: clang static analyzer
      run: cd pappl && make CC=clang
"GHA_ERROR=::error::" clang
    - name: cppcheck
      run: cd pappl && make
"GHA_ERROR=::error::" cppcheck

  build-macos:

    runs-on: macos-latest

    steps:
    - uses: actions/checkout@v2
    - name: install prerequisites
      run: brew install cppcheck libjpeg libpng
libusb
    - name: configure
      run: ./configure --enable-debug --enable-
maintainer --enable-sanitizer
    - name: make
      run: make
    - name: test
      run: make test || cat testsuite/
testpappl.log
    - name: clang static analyzer
      run: cd pappl && make
"GHA_ERROR=::error::" clang
    - name: cppcheck
      run: cd pappl && make
"GHA_ERROR=::error::" cppcheck

  build-windows:

    runs-on: windows-latest

    steps:
    - uses: actions/checkout@v2
    - name: setup-msbuild
      uses: microsoft/setup-msbuild@v1.0.2
    - name: nuget restore
      run: cd vcnet; nuget restore pappl.sln
    - name: msbuild
      run: cd vcnet; msbuild pappl.sln
    - name: test
      run: cd vcnet; ./copy-dlls.bat x64\Debug;
cd x64\Debug; ./testpappl -c -l testpappl.log
-L debug -o testpappl.output -t all
```

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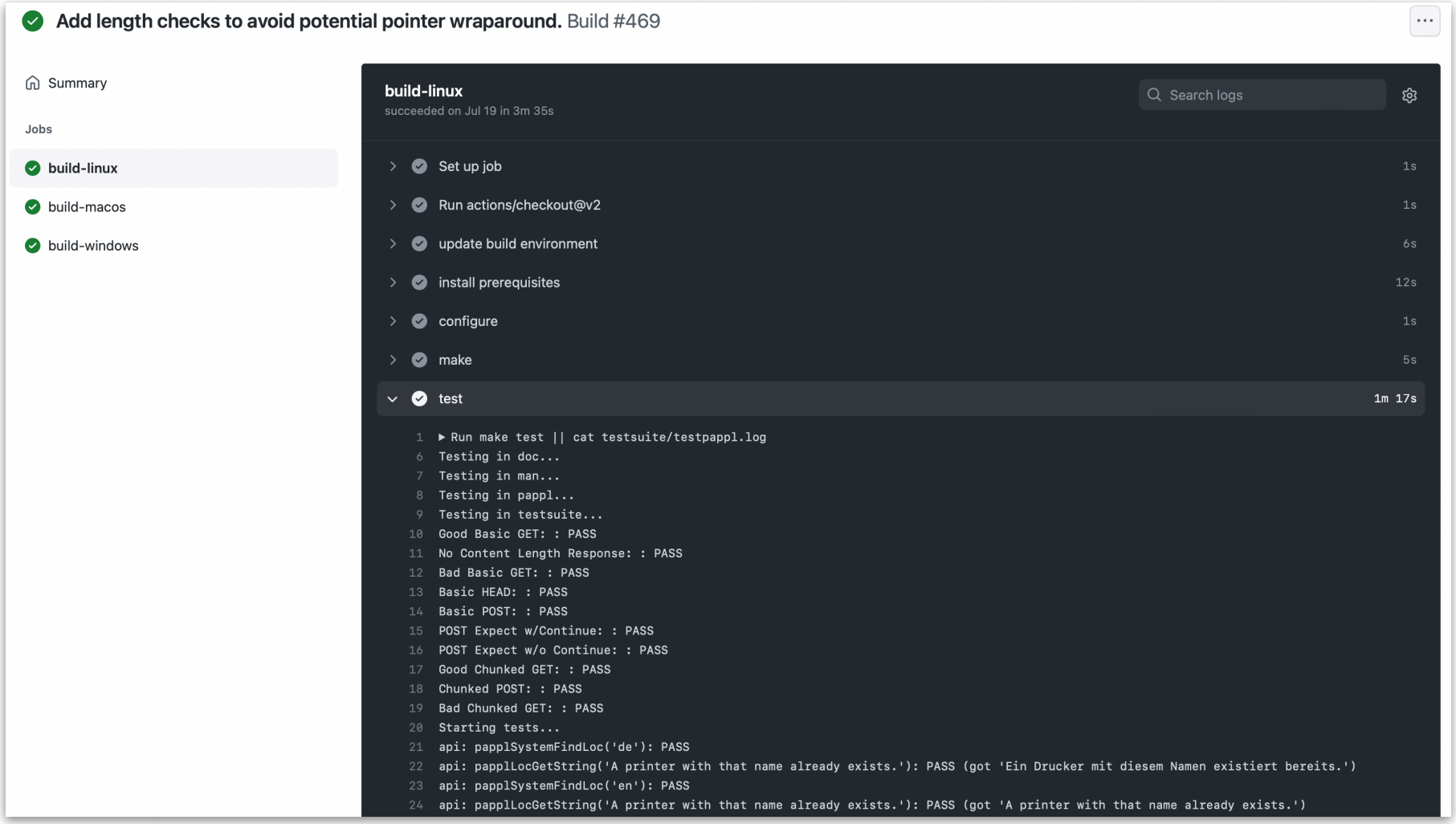
Output: PAPPL build.yml



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Printing Projects Using CI

- CUPS: builders and code analysis, runs full CUPS automated test suite
- ippeveselfcert: builders and code analysis, runs self-cert against ippeveprinter
- ippsample: builders and code analysis, runs various IPP standard test files, transforms, and (work in progress) cloud/proxy tests
- libcups (3.0): builders and code analysis, runs ippeveprinter, ippfind, and ipptool to test basic IPP/2.0 conformance
- LPrint: builders, code analysis, and snapcraft packaging
- PAPPL: builders and code analysis, test suite exercises 90+% of PAPPL API and functionality
- PDFio: builders and code analysis, test suite exercises 100% of PDFio API and functionality

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Resources

- AppImage: https://appimage.org/
- CodeQL: https://codeql.github.com/
- Coverity: https://scan.coverity.com/
- Docker: https://www.docker.com/
- Flatpak: https://www.flatpak.org/
- Github Actions: https://github.com/features/actions
- LGTM: https://lgtm.com/
- Snapcraft: https://snapcraft.io/

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